

CASE 3 — De Beers Tracr: Giving a Physical Diamond a Digital Identity

Company: De Beers Group

Initial development/pilot: 2018

First 100 diamonds tracked: May 2018

Scaled deployment: 2022

Project: Tracr

Sector: Diamonds, luxury products and responsible sourcing

The story

A diamond creates a fascinating engineering problem.

Two stones can look similar, yet their histories may be completely different.

A buyer may therefore want to know:

Where was it mined? Is it natural? Has its identity been substituted? What happened between the mine and the jewellery store?

In January 2018, De Beers announced development of a blockchain initiative designed to create a permanent digital record for registered diamonds.

Then came a remarkable experiment.

On **10 May 2018**, De Beers announced that it had successfully tracked **100 high-value diamonds from mine to retail** using its Tracr blockchain platform. Diamonds were followed from mining through cutting and polishing and onward to jewellery businesses.

Unlike the TradeLens story, this project continued developing.

In **May 2022**, De Beers announced deployment of Tracr at scale. The system combines distributed-ledger technology with security and privacy technologies, and De Beers said it was designed to register up to **one million diamonds per week**.

This is no longer merely an early experiment: De Beers subsequently reported millions of diamonds registered at source.

The engineering concept: Digital Twin

This case introduces students to an extremely important idea:

PHYSICAL OBJECT ↔ DIGITAL IDENTITY

A diamond can be associated with a digital identity containing characteristics such as:

- carat weight;
- colour;
- clarity;
- cut;
- provenance information.

The blockchain then records events concerning that identity as the diamond moves through the value chain.

Tracr also illustrates that blockchain rarely operates alone. De Beers describes a technological combination involving **blockchain, AI, IoT, scanning and advanced security/privacy technologies**.

Focus

Blockchain + Digital Twins + IoT + AI

This is particularly useful for engineering students because the application can be generalized far beyond diamonds:

aircraft components, pharmaceuticals, batteries, luxury watches, industrial machinery, automotive parts.

The engineering issue

There is a fascinating weakness.

The blockchain can tell you that:

Digital Diamond #87543 has a particular history.

But how do you prove that the physical diamond in your hand **really is Digital Diamond #87543?**

This is the **physical-to-digital bridge problem**.

Blockchain secures the digital record; sensors, scanning, certification and identification technologies connect that record to the physical object.

Student task

Your engineering team must adapt the Tracr model to another high-value product: a luxury watch, aircraft component, EV battery or pharmaceutical product.

Questions

1. Why is blockchain useful for proving provenance rather than simply storing product information?
2. How can a physical diamond be reliably connected to its blockchain identity?
3. What roles could AI, IoT sensors and scanning technologies play alongside blockchain?

4. Could a blockchain record remain technically authentic while the physical product associated with it has been replaced or counterfeited? How would you design the system to prevent this?
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